Sumedh Patil

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Experience:

Digital Product Data Analyst - NRB Paints LTD. (India) | June 2025 – Present

- Built and maintained a **SQL-ready dataset for 5,000+ customers**, assigning unique colour codes to custom paint shades and linking customer data with tinting ratios, quantities, and pricing to enable accurate shade reuse and faster repeat orders.
- Integrated datasets with a third-party paint tinting system, validating outputs through controlled adjustments to achieve **>99% accuracy** for new shades and **0% deviation** for repeat orders.
- Implemented quantity-wise and base-specific pricing logic (1L, 5L, 10L, 20L) within the dataset and validated consistency using SQL checks and schema enforcement, delivering a clean, analysis-ready dataset supporting pricing accuracy and trend analysis.

AI/ML Engineering Intern - Rooman Technologies (India) | Sept 2024 – Apr 2025

- Developed and optimized an **end-to-end fraud detection and risk scoring system** using Python, Scikit-learn, and TensorFlow**,** analyzing **280K+ real-world transactions** with highly imbalanced data (<0.2% fraud).
- Trained and evaluated multiple machine learning and deep learning models (Logistic Regression, Random Forest, Neural Networks), improving **ROC-AUC from 0.92 to 0.98** through feature scaling, class weighting, and hyperparameter tuning**.**
- Implemented risk-based probability scoring (0–100) and conducted precision–recall and ROC-AUC analysis to reduce **false negatives by 30%**, aligning ML outcomes with real-world business constraints.

Python Full-Stack Intern - Future Technologys (India) | Mar 2024 – Apr 2024

- Developed strong proficiency in Python, OOP, and core libraries (NumPy, Pandas, Matplotlib, Flask, Scikit-learn), leveraging Django to solve complex problems, manage data (JSON, XML, SQLite), and optimize tasks using multithreading, multiprocessing, and unit testing for efficient, reliable code.
- Created a **weather forecast app with Django**, featuring live **API integration**, dynamic updates, responsive design, interactive displays, and location-based weather predictions.

Technology Stack:

- **Languages**: Python, SQL, Java, C/C++, HTML, CSS, JS, R
- **Libraries**: NumPy, Pandas, Scikit-learn, SciPy, TensorFlow, Matplotlib
- **Frameworks**: Django, Flask, React, FastAPI, PostgreSQL, MySQL, BigQuery, Navicat
- **AI/ML algorithms**: Regression, K-means, Clustering, Random Forest, Predictive Analytics, Decision Tree, KNN
- **Tools**: Azure, Power BI, Tableau, Excel, JIRA, Spreadsheets, Google analytics, Git, GitHub, Microsoft Office, Alteryx, Talend

Education:

- **Bachelors in Computer Science** (**KLS Gogte Institute of Technology,** Belagavi, India) | **Dec 2021 – Jun 2025**
  Courses: Data Structures & Algorithm, Web Dev, Database Management, Computer Networks, Artificial Intelligence and Machine Learning, Applied Statistics, Embedded Systems
- **Pre-University College** (**Govindram Sekseria Science College,** Belagavi, India) | **June 2019 – Jul 2021**
  Courses: Physics, Mathematics, Chemistry, Computer Science

Projects:

Product Demand Forecasting & Inventory Intelligence System **| June 2025 – Aug 2025**

- Architected an end-to-end data and analytics pipeline using Python (Pandas, Faker), PostgreSQL, and SQL to generate and analyze **5,000+ synthetic retail transactions**, performing category-level and time-series demand analysis to support inventory planning without relying on confidential data.
- Applied regression-based demand forecasting with time-based feature engineering and proper train–test splits, evaluating accuracy using **Mean Absolute Error (MAE)**, and **exposed insights via FastAPI REST APIs and interactive Tableau dashboards**, enabling production-style analytics delivery and business visibility, improving end-to-end visibility and workflow maintainability.

Fraud Detection and Risk Scoring Analysis **| Dec 2024 – Feb 2025**

- Developed a transaction-level fraud detection model using **Python and Scikit-learn on 280K+ financial records**, focusing on risk identification in highly imbalanced datasets typical of banking and payments systems.
- Applied feature engineering, data validation, and imbalance handling (class weighting / SMOTE) to improve fraud capture, prioritizing recall and risk sensitivity to reduce false negatives in high-impact transactions.
- Assessed model effectiveness using precision, recall, F1-score, and ROC-AUC, delivering a transparent, explainable baseline suitable for risk monitoring and control frameworks in regulated financial environments.

Weather forecast application using Django **| April 2024**

- Engineered an end-to-end data pipeline using Python and Django to ingest, validate, and store real-time weather data from external REST APIs, processing 10K+ records/month across global locations with structured transformation for analytical use.
- Performed data preprocessing, feature extraction, and exploratory data analysis (EDA) on weather parameters (temperature, humidity, pressure, wind speed), enabling trend analysis and city-level comparisons with 99% data accuracy and consistent schema enforcement.
- Optimized backend data workflows using efficient API calls, modular services, and caching, reducing data ingestion latency by 25%, improving query performance, and laying the foundation for ML-ready datasets and future predictive modelling.